Titanic Dataset Exploratory Data Analysis (EDA) Report

(Week 1 of internship with Analyst Lab Africa.)

By: IBINABO DIREH

Date: MAY, 2026

1. Introduction

The purpose of this project was to perform an Exploratory Data Analysis (EDA) on the Titanic dataset. The analysis was carried out to understand passenger characteristics and how they were related to survival, and prepare the dataset for future machine learning tasks.

The dataset contains information about passengers aboard the Titanic, including demographic details, ticket information, and survival outcomes.

2. **Dataset Overview**

Feature	Description
PassengerId	Unique passenger identifier
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Passenger class (1 = 1st, 2 = 2nd, 3 = 3rd)
Name	Passenger name
Sex	Passenger gender
Age	Passenger age
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Ticket	Ticket number
Fare	Ticket fare
Cabin	Cabin number
Embarked	Port of embarkation

The dataset contains 891 records and 12 features.

3. Data Cleaning and Preparation

Some data quality issues were identified and resolved.

Missing Values

* The Age column contained missing values and was filled using the mean.
* The Cabin column contained a large number of missing values and was removed from the dataset.
* Rows with missing Embarked values were dropped due to their small number.

Data Transformation

* Categorical variables were encoded to make the dataset suitable for future machine learning applications.

4. Exploratory Data Analysis

Passenger Demographics

Passengers belonged to different age groups, genders, and travel classes. Most passengers were third class travelers, while first-class passengers were the smallest group.

Survival by Gender

Survival rates differed greatly between male and female passengers. Female passengers were more likely to survive compared to male passengers (despite there being more males).

This suggests that rescue efforts mostly likely prioritized women during the disaster.

Survival by Passenger Class

Passenger class had a noticeable impact on survival.

* First-class passengers had the highest survival rates.
* Second-class passengers showed moderate survival rates.
* Third-class passengers had the lowest survival rates.

This indicates that social and physical factors related to passenger class most likely influenced survival outcomes.

Age Distribution

Passenger ages were of a wide range from children to elderly adults. Younger passengers appeared to have slightly better survival chances, although the relationship was not as clear as observed for gender and passenger class.

Fare Analysis

Ticket fares varied considerably across passengers. Higher fare generally corresponded to higher passenger class and increased survival chances.

5. Key Findings

The analysis revealed several important patterns:

1. Gender was one of the strongest factors associated with survival.
2. First-class passengers had higher survival rates than second and third-class passengers.
3. Higher ticket fares were generally associated with better survival outcomes.
4. Age showed some relationship with survival, although not as clearly as with gender and passenger class.

6. Assumptions and Limitations

Assumptions

* Missing Age values were assumed to be adequately represented through imputation of mean.
* Rows with valued missing from Embarked column were assumed to have little or no impact on the overall analysis.
* The Survived column was assumed to contain accurate labels.

Limitations

* Some features contained missing values that may affect interpretation.
* Cabin information was largely missing, therefore excluded from analysis.
* No statistical significance testing was performed.
* The analysis focused on identifying patterns rather than building learning models.

7**. Conclusion**

This exploratory analysis revealed valuable insights into factors associated with passenger survival on the Titanic. Gender and passenger class emerged as the most important characteristics, while fare and age also showed meaningful relationships with survival outcomes. The cleaned dataset is ready for making machine learning models.

I experienced some difficulty documenting on GitHub for the first time. I also had some difficulty meeting up with deadline and aim to improve on that in subsequent weeks.

This project helped me gain an introductory experience in data analysis.

Thank you.